FINAL COMPLETE NOTES — STATISTICS & HYPOTHESIS TESTING

Scientific Temper

Scientific Temper means having a logical, questioning, and evidence-based way of thinking instead of believing things blindly.

Main Features:
• Logical thinking
• Curiosity
• Observation
• Experimentation
• Open-mindedness
• Decision based on evidence

In Short:
Scientific temper is the habit of thinking scientifically and solving problems using logic and evidence.

What is Hypothesis Testing?

Hypothesis Testing is a statistical method used to check whether an assumption about data is true or false.

It helps decide:
• Result real hai
• Ya random chance ki wajah se aya hai

H0 and H1

H0 (Null Hypothesis):
• No change
• No effect
• Default condition

H1 / Ha (Alternative Hypothesis):
• Change exists
• Effect exists

Example:
H0: This medicine does NOT cure fever.
H1: This medicine cures fever.

P-Value and Significance Level

P-value batati hai:
“Observed result random chance se aane ki probability kitni hai.”

Usually:

$\alpha = 0.05$

Decision Rule:
• $p\text{-value} < 0.05 \rightarrow$ Reject H0
• $p\text{-value} > 0.05 \rightarrow$ Do not Reject H0

Website Conversion Example

Old Website:
100 sales out of 1000 visitors = 10%

New Website:
125 sales out of 1000 visitors = 12.5%

Increase:
25%

Hypothesis testing check karta hai:
Real improvement hai ya random chance?

Types of Errors

Type I Error:
• Also called False Positive / Alpha Error
• Problem nahi thi but system ne problem bol diya

Example:
Healthy person ko sick bol diya.

Type II Error:
• Also called False Negative / Beta Error
• Problem actually present thi but system bolta problem nahi hai

Example:
Cancer patient ko healthy bol diya.

Which Error is More Dangerous?

Depends on situation.

Type II Error dangerous in:
• Cancer detection
• Fraud detection
• Fire alarm
• Security systems

Type I Error dangerous in:
• Court judgement
• Spam filters
• Bank transaction blocking

One Tail and Two Tail Test

One Tail Test:
Used when sirf ek direction check karni ho.

Example:
New method better hai?

Two Tail Test:
Used when any difference check karna ho.

Example:
New method different hai?

Confidence Interval

95% Confidence Interval means:
“Hume 95% confidence hai ki actual value given range me hogi.”

God Example

H0: God is not present
H1: God is present

Possible Outcomes:
• God not present + “God not present” → Correct
• God not present + “God present” → Type I Error
• God present + “God present” → Correct
• God present + “God not present” → Type II Error

Memory Tricks

• Small p-value → Reject H0
• Large p-value → Reject mat karo
• Type I → False alarm
• Type II → Real danger miss
• Type I → False Positive / Alpha Error
• Type II → False Negative / Beta Error

Quick Revision Table

Term	Meaning
H0	No effect / No difference
H1	Effect exists / Difference exists
Type I Error	False Positive
Type II Error	False Negative

p-value < 0.05	Reject H0